ARCHITECTURE AS RELATIONAL STRUCTURE

From Geometry to Topology: Graph Thinking in Design

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A CITY IS NOT A TREE

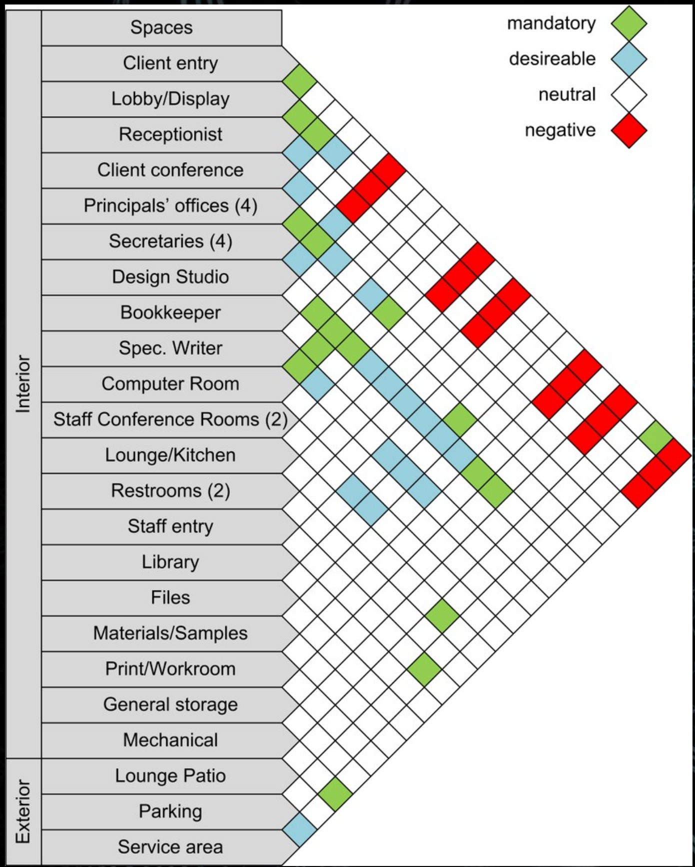
"A tree is a structure which is not sophisticated enough to capture the complex social and physical reality of a city... The city is not a tree; it is a ***semilattice***."

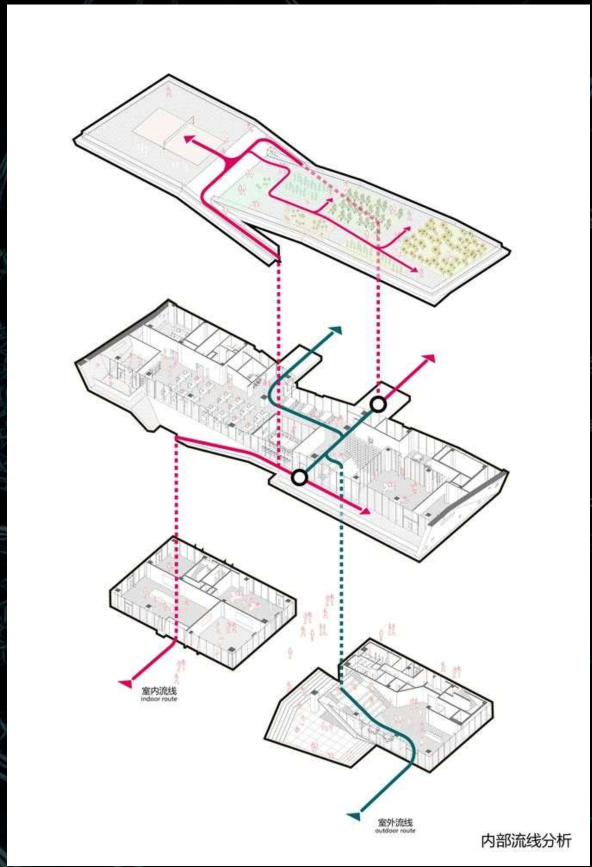
— Christopher Alexander (1965)

A BUILDING IS NOT A TREE

HOW IS ARCHITECTURE RELATIONAL?

- Architecture is not only objects, but relationships between spaces
- Movement, visibility, adjacency, hierarchy → inherently relational
- Traditional representation (plans, sections) obscures relational structure
- Graphs make relationships explicit, computable, analysable
- Hypothesis: Buildings are better understood as networks of spatial relations than as collections of geometric primitives


Spaces
Client entry
Lobby/Display
Receptionist
Client conference
Principals' offices (4)
Secretaries (4)
Design Studio
Bookkeeper
Spec. Writer
Computer Room
Staff Conference Rooms (2)
Lounge/Kitchen
Restrooms (2)
Staff entry
Library
Files
Materials/Samples
Print/Workroom
General storage
Mechanical
Lounge Patio
Parking
Service area
Interior
Exterior
mandatory
desireable
neutral
negative


室内流线
indoor route
室外流线
outdoor route
内部流线分析


level 2
level 1
level 0
level -1
1
2
3
4
5
6

FROM GEOMETRY TO TOPOLOGY

- Geometry: coordinates, distances, shapes
- Topology: connectivity, adjacency, continuity (invariant under deformation)

Concept	Geometry	Topology
Focus	Shape & size	Relationships
Changes under deformation	Yes	No
Example	Wall length	Room adjacency

WHAT ARE GRAPHS

"A graph is an object consisting of two sets called its vertex set and its edge set. The vertex set is a finite nonempty set. The edge set may be empty, but otherwise its elements are two-element subsets of the vertex set." R.J. Trudeau, Introduction to Graph Theory

GRAPH

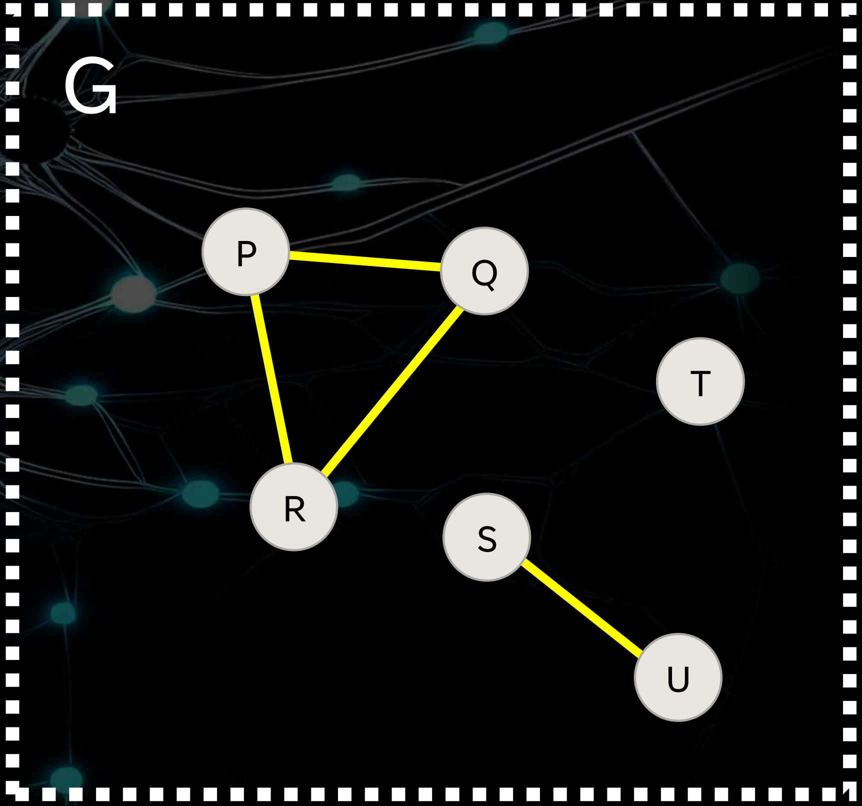
VERTEX SET

$\{P, Q, R, S, T, U\}$

EDGE SET

$\{\{P,Q\}, \{P,R\}, \{Q,R\}, \{S,U\}\}$

Attributes
Vertex properties: area, function, occupancy
Edge properties: distance, permeability, visibility

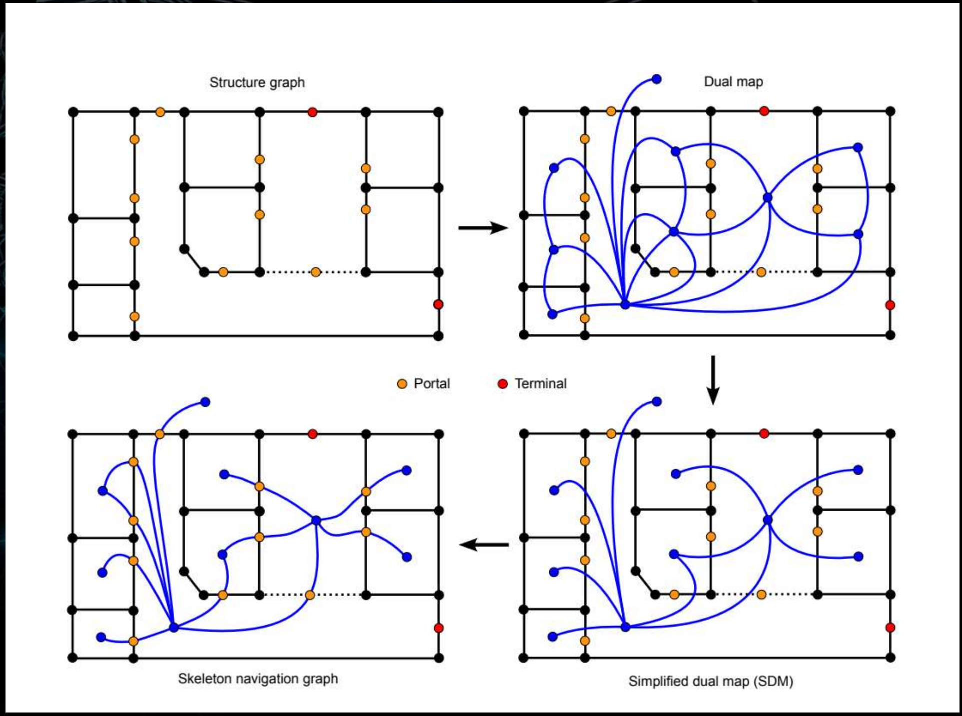


GRAPHS RELEVANT TO AEC:

- Adjacency graphs
- Connectivity graphs
- Visibility graphs (VGA)
- Circulation graphs

REPRESENTING BUILDINGS AS GRAPHS

- Room → node
- Door/shared boundary → edge
- Variants:
  - Centroid-based graphs (dual graphs)
  - Grid-based graphs (VGA)
  - Line-based graphs (axial maps)
- Important Distinction:
  - Primal vs Dual representations
  - Choice depends on analysis goal



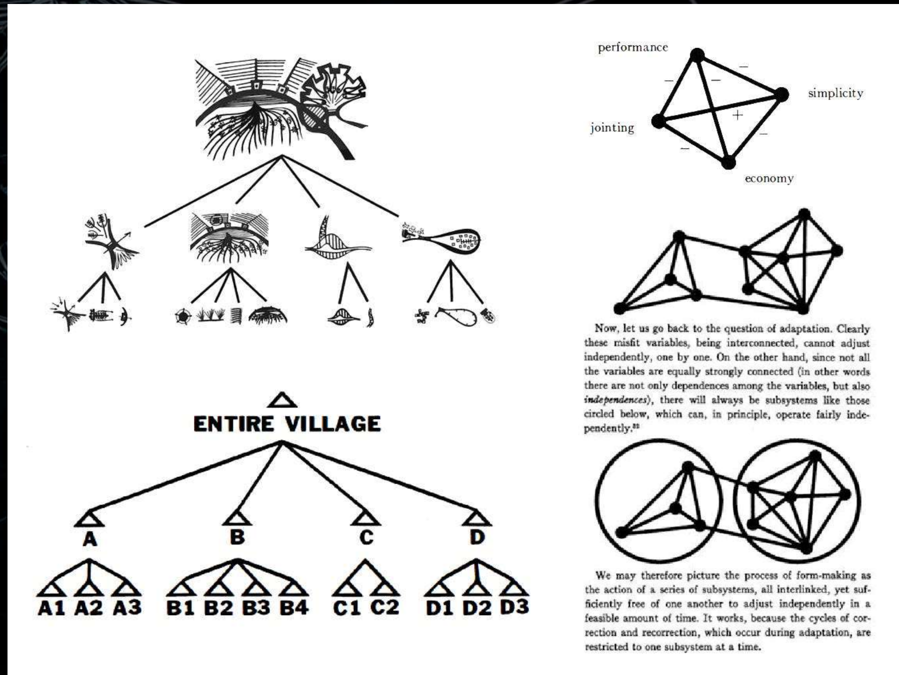
THE ANALYTICAL POWER OF GRAPHS

- Moves architecture toward quantitative reasoning
- Examples:
  - Shortest path (navigation)
  - Centrality (importance of spaces)
  - Accessibility (space syntax)
  - Clustering (functional zoning)


High
Low
Main Entrance
Secondary Entrance

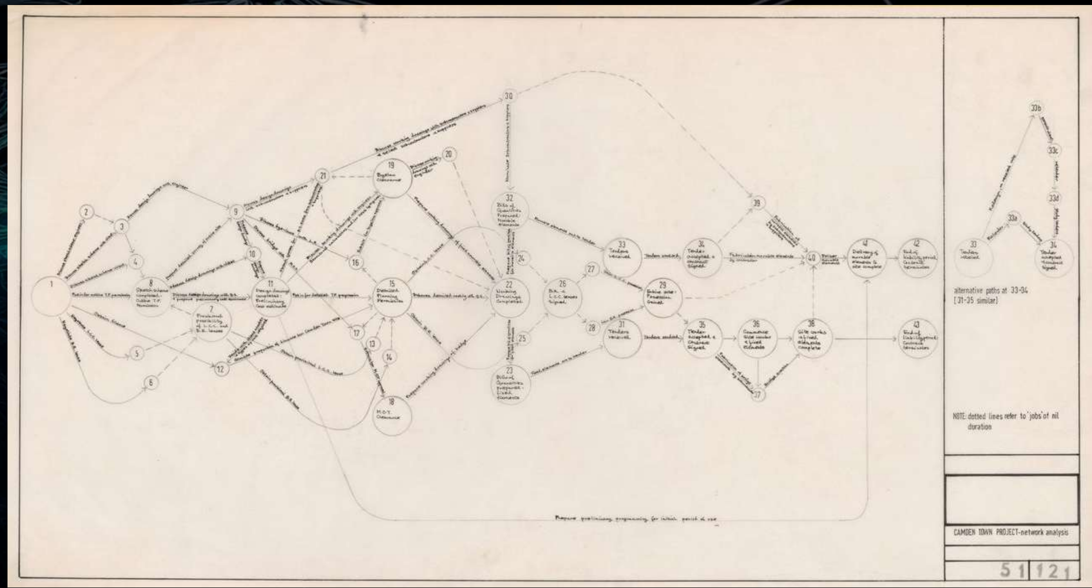
HISTORICAL FOUNDATIONS: ARCHITECTURE AS RELATIONAL STRUCTURE – KEY FIGURES

- Christopher Alexander
  - Notes on the Synthesis of Form (1964)
    - Semi-lattice vs tree structures
    - Argued cities are complex relational systems



HISTORICAL FOUNDATIONS: ARCHITECTURE AS RELATIONAL STRUCTURE – KEY FIGURES

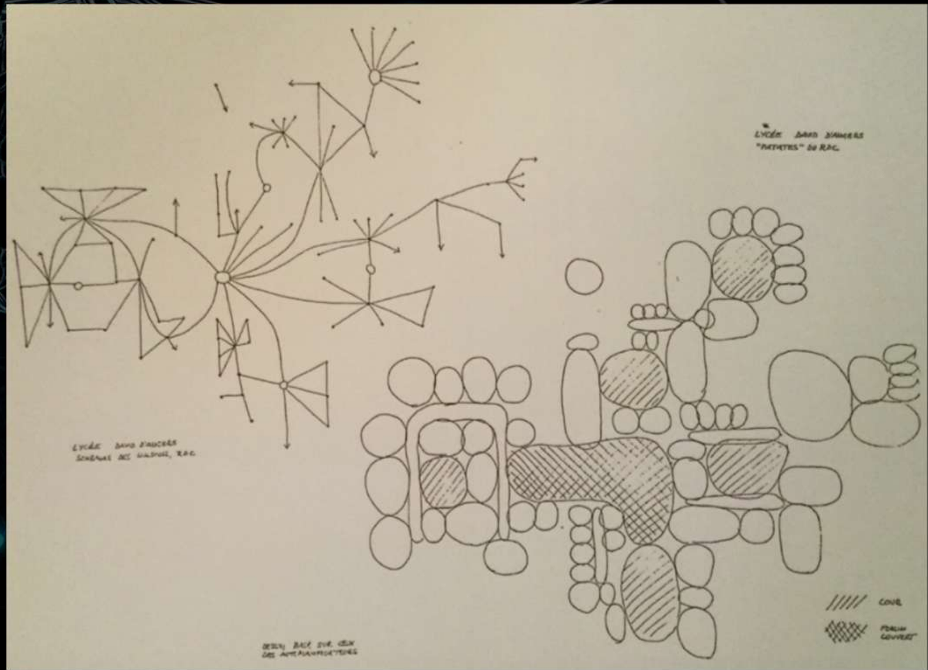
- Cedric Price
  - Fun Palace (1964)
    - Architecture as dynamic system
    - Programmatic relationships over form



Fun Palace: network analysis

HISTORICAL FOUNDATIONS: ARCHITECTURE AS RELATIONAL STRUCTURE – KEY FIGURES

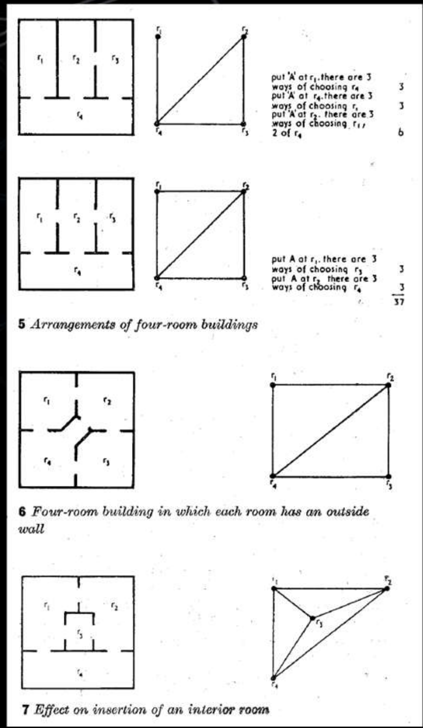
- Yona Friedman
  - Lycée David d’Angers (1978)
    - Spatial networks and flexible infrastructures
    - Represented architecture through mathematical diagrams of discrete elements, such as rooms, and their relationships.
    - His graphs also showed how people moved through a space



Yona Friedman, Lycée David d’Angers, « Patates » du rez-de-chaussée, 1980

HISTORICAL FOUNDATIONS: ARCHITECTURE AS RELATIONAL STRUCTURE – KEY FIGURES

- Peter H. Levin
  - “Use of Graphs to Decide the Optimum Layout of Buildings” Architect’s Journal (1964)
    - Pioneered the use of graph theory for architectural space planning
    - Introduced methods to determine optimal room layouts based on required adjacencies, transforming relationship matrices into planar graphs to minimize circulation costs or optimize functional proximity.



5 *Arrangements of four-room buildings*

6 *Four-room building in which each room has an outside wall*

7 *Effect on insertion of an interior room*

HISTORICAL FOUNDATIONS: ARCHITECTURE AS RELATIONAL STRUCTURE – KEY FIGURES

- B. Whitehead and M.Z. Eldars
  - “The Planning of Single-Storey Layouts”, Building Science (1965)
    - Addressed the optimization of floor plans by focusing on the efficiency of internal circulation, traffic, and movement between rooms, specifically for single-storey buildings.
    - Developed a systematic approach for automated layout design, which involves identifying the optimal location for rooms or facilities based on frequency of trips and transportation costs.

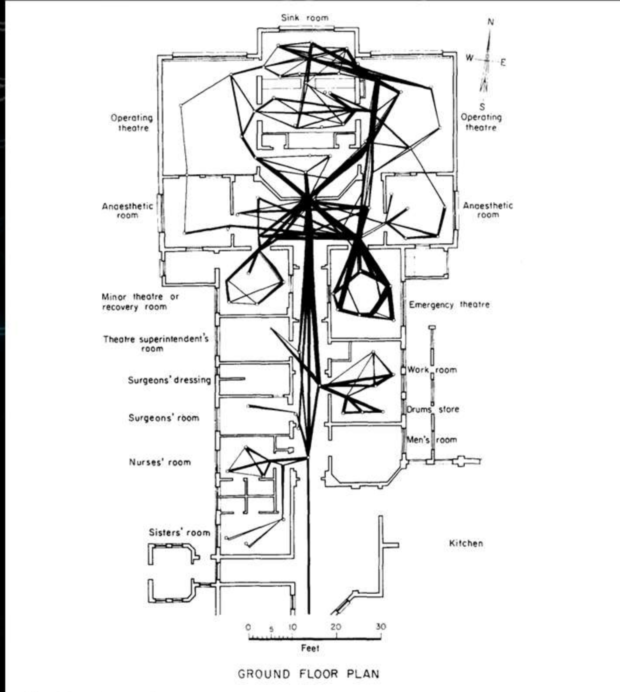
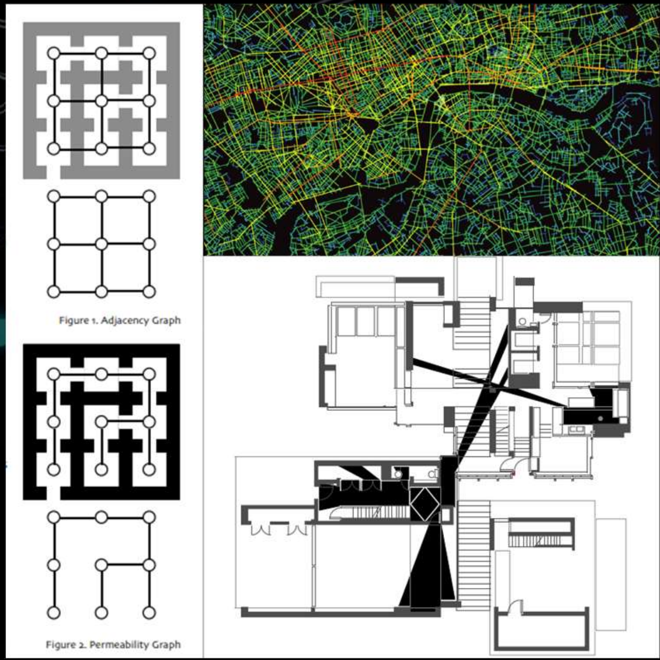


Fig. 1. String diagram of movements of state enrolled nurse during one day's duty in existing operating theatre suite.

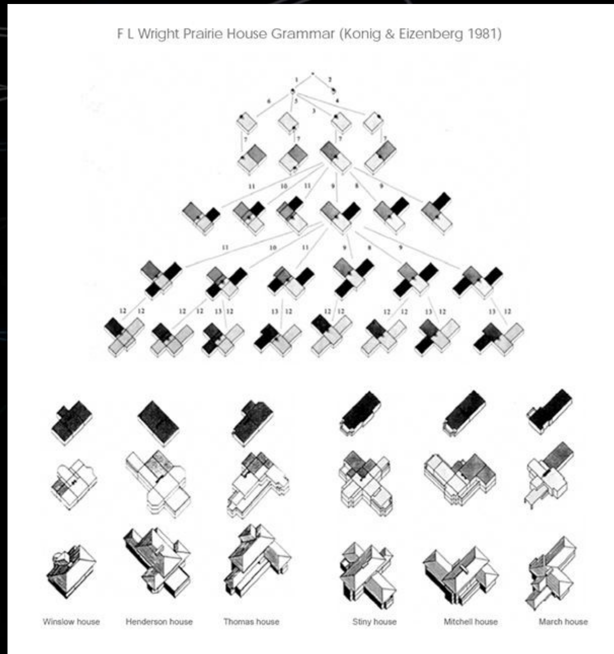
SPACE SYNTAX (1970S -)

- Bill Hillier, Julienne Hanson, John Peponis, and others
  - Social Logic of Space: Developed at University College London in the 1970s, space syntax is based on the theory that spatial configuration is intrinsic to human activity. It posits that the way buildings and streets are laid out directly influences social interaction, economic value, and patterns of movement.
  - Quantitative Network Analysis: It uses graph theory to represent architectural or urban environments as networks of nodes and links. Key analytical measures include Integration (how accessible a space is from all others) and Choice (the likelihood of a space being used as a through-route), which help predict where people will naturally walk, stand, or interact.
  - Evidence-Based Design Tool: Beyond academic theory, space syntax provides a science-based modeling approach used by Space Syntax Limited to forecast the impact of urban design decisions. High-profile applications include the redesign of Trafalgar Square and the optimization of layouts for major hospitals, museums, and entire cities



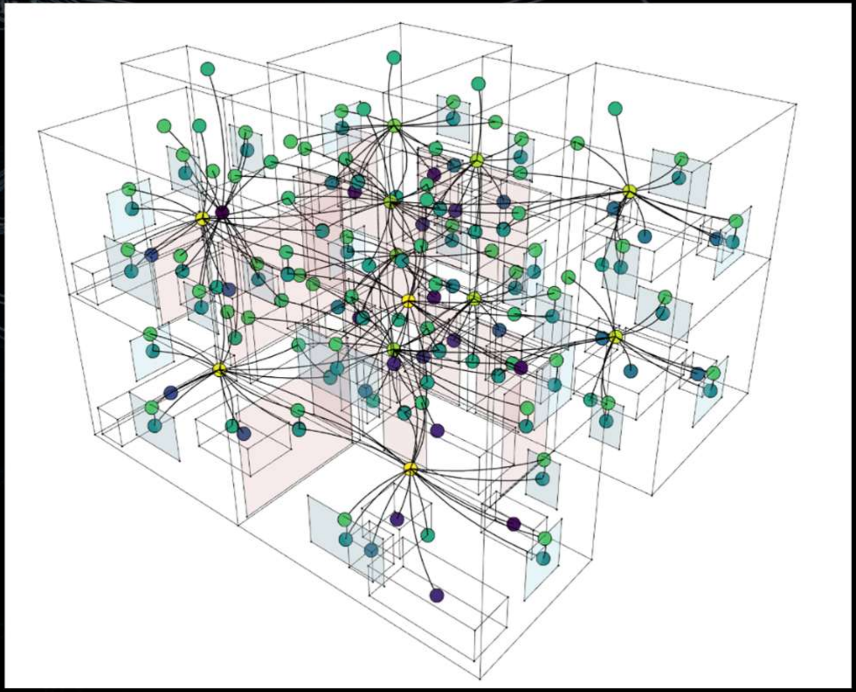
COMPUTATIONAL DESIGN & GRAPH THINKING (1990S–2010S)

- George Stiny: Rule-based generative systems
- Charles Eastman: Building Information Modelling
- William J. Mitchell: Digital design + relational modelling (Non-manifolds)
- Michael Batty: Cities as networks and complex systems



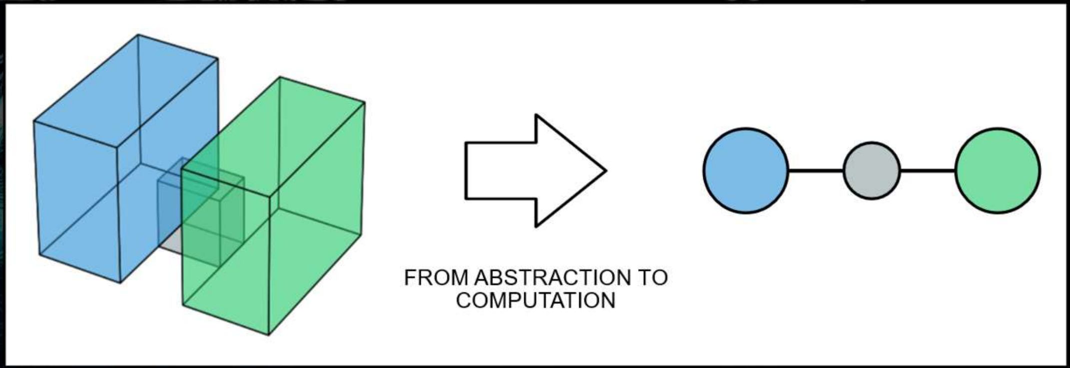
GRAPH-BASED AEC & URBAN SYSTEMS (2010S–PRESENT)

- Key Directions
  - BIM → relational databases → graph databases
  - Emergence of: Neo4j and Kuzu
- Recent Developments
  - Graph Machine Learning (GML)
  - GraphRAG-style generative workflows
  - Spatial-semantic integration (e.g., TopologicPy)



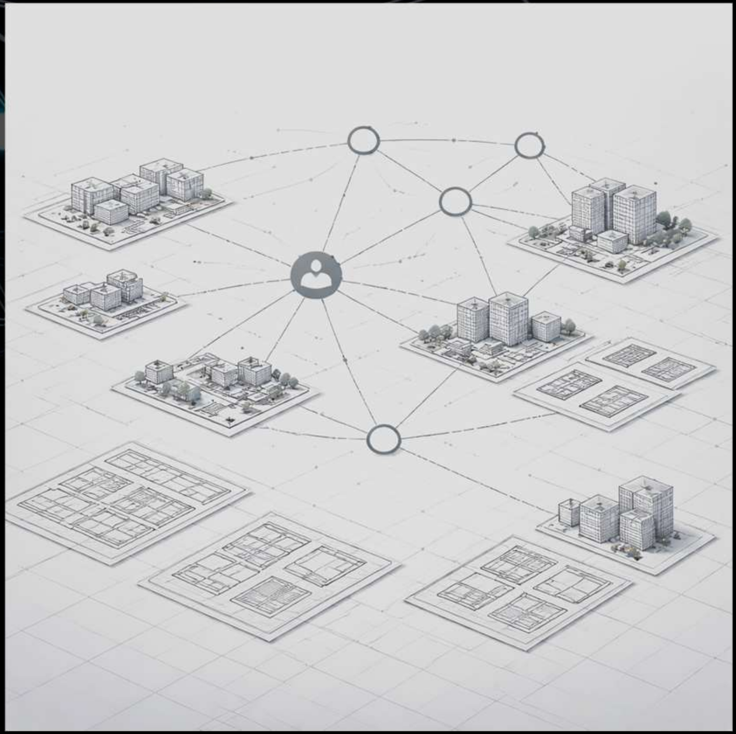
CONCEPTUAL SYNTHESIS

- Geometry is representation
- Graphs are structure
- Both are abstractions
- Both are open to computation



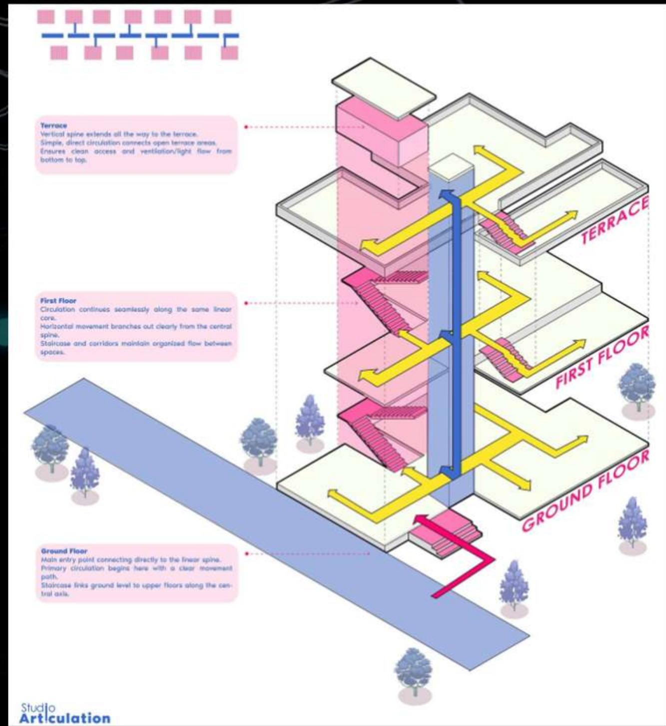
IMPLICATIONS FOR AEC PRACTICE

- Design Thinking Shift
  - From objects → relationships
  - From form-making → system structuring
- Interoperability & Data Models
  - IFC already encodes relationships
    - Graphs provide:
      - clearer semantics
      - easier querying
      - better extensibility



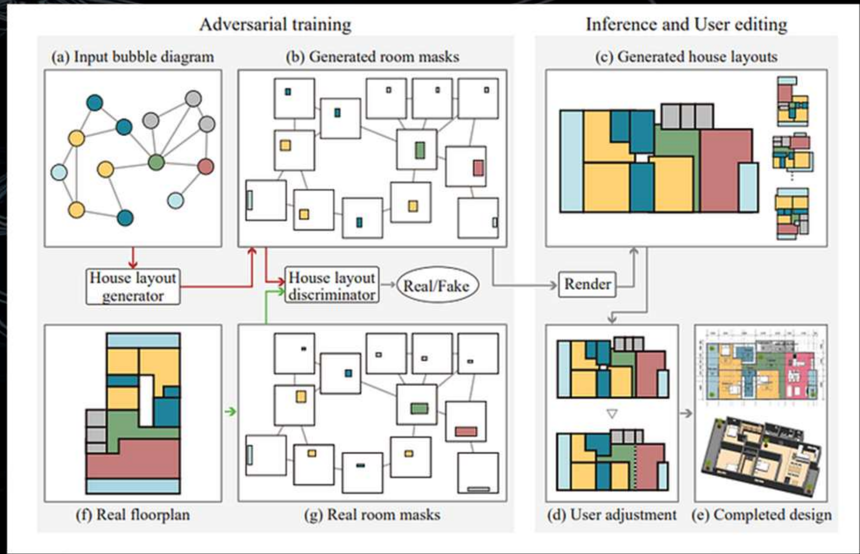
IMPLICATIONS FOR AEC PRACTICE

- Performance Analysis
  - Circulation efficiency
  - Energy modelling (via relational zones)
  - Visibility and safety



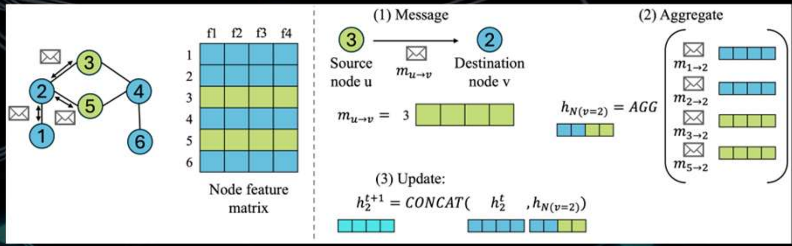
IMPLICATIONS FOR AEC PRACTICE

- Generative Design
  - Graph-based generation:
    - constraint-driven layouts
    - precedent-informed synthesis
  - GraphRAG-like workflows:
    - retrieve spatial patterns
    - generate new layouts with constraints



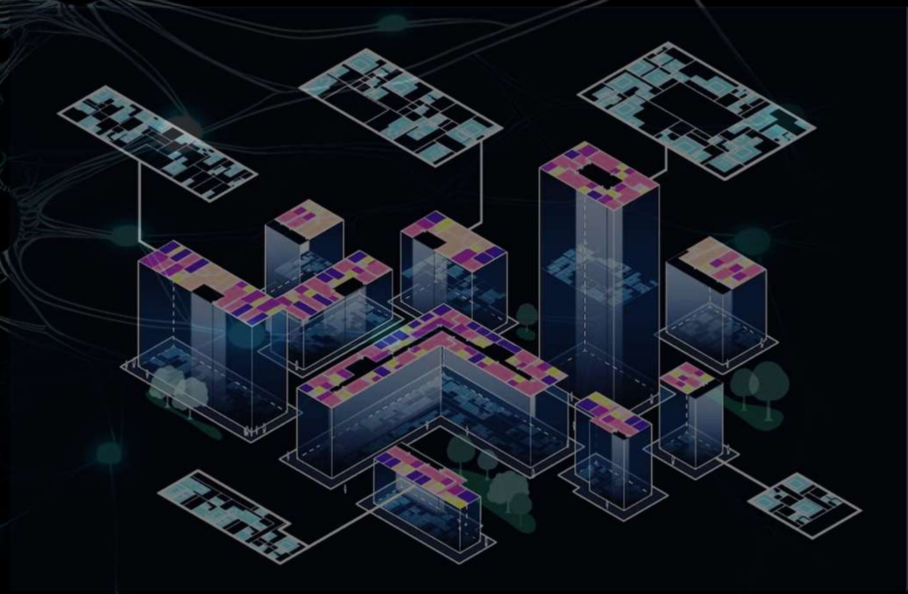
IMPLICATIONS FOR AEC PRACTICE

- AI + Architecture
  - Graphs are the native structure for:
    - Graph Neural Networks (GNNs)
    - Relational reasoning
    - Spatial prediction



THE FUTURE

- The future of architectural computation is not geometric-first, but relational-first
- Graphs unify:
  - Geometry
  - Topology
  - Semantics
  - They enable analysis, simulation, generation



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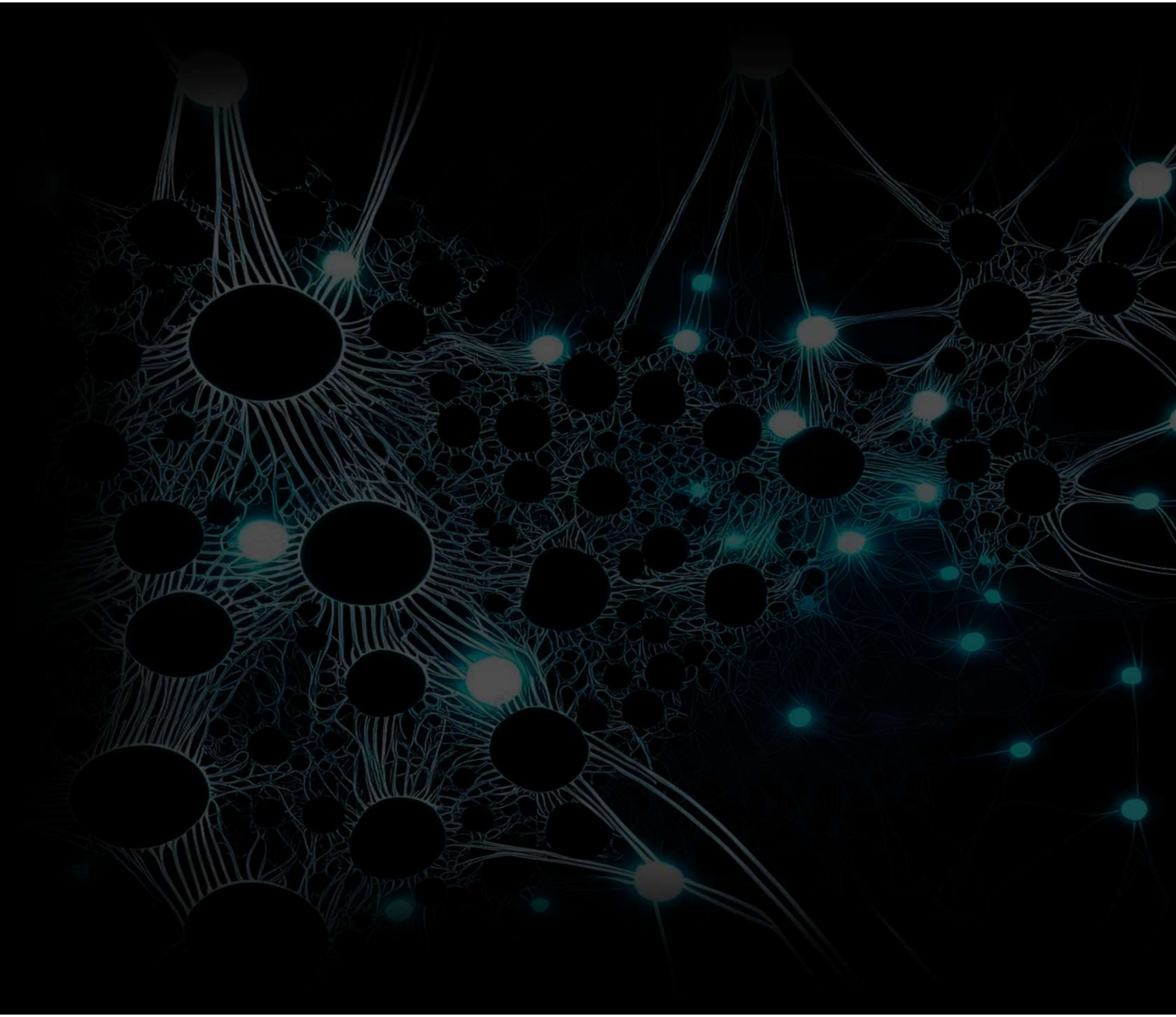

THANK YOU

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